0625 | PAPER 4

TOPIC 03
Waves

59 topical answers • 2021–2025
Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme.
For drawings and graph lines, compare the stated method with the original question figure.

Topic contents

Core formulas

- wave speed $v = f\lambda$
- refractive index $n = \sin i \div \sin r$
- critical angle: $n = 1 \div \sin c$
- echo distance = speed × time ÷ 2
- magnification = image height ÷ object height

3.1 Wave properties

Q127 • May/June 2022 Paper 42 Question 5 — 3.1 Wave properties

(a)(i) ultrasound or sound (frequency) above audible range

frequency > 20 kHz or 20 000 Hz

(a)(ii) 8.7 × 10-4 m

(λ =) v / f or v = fλ in any form

(λ =) 1.3 × 103 / 1.5 × 106 or 8.7 × 10n

(b) basic description of use e.g. X-rays for detecting broken bones

additional detail e.g. X-rays pass through soft tissue and not through bone

Q128 • Oct/Nov 2022 Paper 43 Question 6 — 3.1 Wave properties

(a)(i) correct direction, with angle made with surface correct

(a)(ii) three wavefronts perpendicular to their answer to (a)(i)

wavelength 1.6 cm / same as incident wave

(b) at least two correct arcs (after the gap in the barrier)

three circular arcs (after the gap on the barrier) centred on gap

wavelength same as wavelength of incident wavefronts

(c) 6.4 cm / s or 0.064 m / s

v = fλ or (v =) fλ or 4.0 × 1.6 or 4.0 × 0.016

Q129 • May/June 2023 Paper 42 Question 6 — 3.1 Wave properties

(a) P-waves: longitudinal

S-waves: transverse

(b) 1600 m or 1.6 km

v = fλ or (λ =) v / f

(λ =) [7.2 × 1000] / 4.5 or (λ =) 1.6 × 10N or (λ =) 7.2 / 4.5

Q130 • Oct/Nov 2023 Paper 43 Question 5 — 3.1 Wave properties

(a) 3 wavecrests same wavelength as original

3 arcs

3 semi-circular wavecrests centred on middle of gap

(b) 3 parallel wavecrests on left, reduced wavelength

angular spread must be less / reduced divergence

3 wavecrests on right with reduced curvature compared with (a)

no reverse curvature anywhere

Q131 • May/June 2024 Paper 41 Question 5 — 3.1 Wave properties

(a) Any two valid point(s):

- (longitudinal) vibration / oscillation in wave parallel to propagation direction / direction of travel
- transverse wave vibrates / oscillates perpendicular to propagation direction / direction of travel
- (longitudinal) consists of compressions and rarefactions
- transverse wave consists of crests / peaks and troughs
- (longitudinal) needs a medium (to travel)

(b)(i) P-wave and it is longitudinal

(b)(ii) 1.8 × 104 m or 18 km

1.5λ or 1.5 × 1.2 × 104 or 1.8 × 10N

(b)(iii) v = f λ or v= λ / t or (t =) λ / v or f = 4600 / 1.2 × 10 4

or (t / 5 =) 1.2 × 104 / 4600 or (t =) 6(.0) × 104 / 4600

13 s

t = 1 / f or (time for one wave =) 2.6 (s)

Q132 • Oct/Nov 2024 Paper 43 Question 9 — 3.1 Wave properties

(a) Q and V

(b) red shift

(galaxy) moving away / receding (from Earth/us)

(c) galaxies further away (are receding) with higher speed or galaxies further away have greater redshift

Universe is expanding

3.2 Reflection of waves

Q133 • May/June 2022 Paper 43 Question 7 — 3.2 Reflection of waves

(a) ray from left hand corner of the mirror to the eye

angle of incidence = angle of reflection

(b) Any two valid point(s):

virtual

upright

same size as object

laterally inverted

(c)(i) ultraviolet / X-rays / gamma rays

(c)(ii) infrared / microwaves / radio (waves)

Q134 • Oct/Nov 2023 Paper 42 Question 5 — 3.2 Reflection of waves

(a) indication of position of car along a straight line from X above and to left of road at junction.

(b) Incident ray from car to mirror and reflected ray from mirror towards X

angle of incidence equal to angle of reflection

(c) converging lens (to left of eye)

rays refracted by additional converging lens

rays refracted by lens in eye to give converging rays

focal point of rays / image on retina

3.3 Sound and ultrasound

Q135 • Feb/March 2021 Paper 42 Question 5 — 3.3 Sound and ultrasound

(a) echo

(b) (λ=) 7.5 × 10-4 m

(λ=) v / f or v = fλ in any form

(λ=) 1.5 × 103 / 2 × 106

(c)(i) ,(ii) labelled wavelength of incident wave

3 part circles to the left of the barrier and centred to right of the barrier

wavelengths of reflected and incident waves same

Q136 • May/June 2021 Paper 41 Question 6 — 3.3 Sound and ultrasound

(a) two points labelled C at the centre of the two compressions

(b) 6200-6500 Hz

(λ =) value from 0.051 to 0.053 (m) seen anywhere

(f =) v / λ in any form or 330 / 0.052 or 330 / 5.2 or 63

(c) compressions / rarefactions closer or more compressions / rarefactions (in same distance)

less diffraction / spreading out

(because of) smaller wavelength or ratio wavelength / gap width smaller

Q137 • May/June 2021 Paper 43 Question 5 — 3.3 Sound and ultrasound

(a)(i) part of a circle, at least quarter of a circle, centred on centre of gap

waves same wavelength as incident waves

(a)(ii) waves pass through gap remaining straight

less / no diffraction occurs

(b) 1.8 m

$\lambda = v/f$ or 1500/850 in any form

Q138 • Oct/Nov 2021 Paper 42 Question 6 — 3.3 Sound and ultrasound

(a)(i) C in line with smallest gap between dots

(a)(ii) R in line with largest gap between dots

(a)(iii) arrow corresponds to wavelength

(b) 1500 m / s

(c) $v = f\lambda$ in any form or ($f =$) v / λ

($f =$) 1500 / 0.12

($f =$) 13 kHz or 13 000 Hz

(d) statement consistent with candidate's answer to 6c

ultrasound is above 20 000 Hz

Q139 • Oct/Nov 2021 Paper 43 Question 6 — 3.3 Sound and ultrasound

(a)

method of producing sound, e.g. clap for echo method; or gun for direct measurement, sig gen

on block

apparatus used, e.g. stopwatch, long tape, trundle wheel, wall if using echo method, metre rule, microphones and timer or

microphones and oscilloscope

detail of measurement of (long) distance, e.g. measure distance between person and the wall, measure distance between

loudspeaker and microphone or measure distance between two microphones

detail of measurement of time; or appropriate time measured, e.g. at one end start stopwatch when smoke seen from

Key physics: gun stop sound heard start stopwatch / clap echo measure time

taken between clap and hearing echo, timer starts when first microphone receives signal and stops when second receives

signal; or measurement of wavelength, e.g. move one microphone away until two waves on oscilloscope have moved

one wavelength apart

speed = measured distance / time for direct method

or speed = 2 × distance from student clapping to wall / time for echo method

or distance between microphones = wavelength and $v = f \times \lambda$

(b)

wavelength of light is (much) smaller than width of doorway or wavelength of sound

wavelength of sound is similar to width of doorway; or $\lambda \simeq$ width of gap for diffraction to occur

results in greater diffraction

Q140 • May/June 2022 Paper 43 Question 6 — 3.3 Sound and ultrasound

(a)(i) wavefronts semicircles or part semicircles centred on gap

wavelength of waves to right of barrier same as wavelength of incident wave

(a)(ii) 1 wavelength shorter

correct refraction

2 direction of travel perpendicular to wavefronts

(b) Any two valid point(s):

• particles (in transverse waves) vibrate perpendicular to the direction of travel (of the wave) or

particles in longitudinal waves vibrate parallel to the direction of travel of the wave

• longitudinal waves have compressions and rarefactions
• transverse waves have troughs and crests
(c)(i) 1000 m / s ≤ value ≤ 2000 m / s
(c)(ii) molecules closer together / water has greater density

Q141 • Feb/March 2023 Paper 42 Question 6 — 3.3 Sound and ultrasound

(a) (region where) particles are close(r) together (than normal); or (region where) there is a great(er) pressure (than normal)
(region where) particles are further / far apart (than normal); or (region where) there is a low(er) pressure (than normal)
(b) Key physics: light does not need medium travel through sound needs no
between Sun and Earth)
(c) 3100 m or 3.1 km
v = s / t or (s =) vt or 340 × 9
(d) 1400 C
I = Q / t or (Q =)I t or 3.0 × 104 × 48 × 10-3
(t =) 48 × 10-3 or (t =) 4.8 × 10-2 or (t =) 0.048 SEEN

Q142 • May/June 2024 Paper 42 Question 6 — 3.3 Sound and ultrasound

(a)(i) C marked and labelled at a peak of the sine wave
R marked and labelled at a trough of the sine wave
(a)(ii) graph / it does not show (variation with) displacement
(a)(iii) (amplitude) increases and (frequency) decreases
(b) 0.12 m
v = f λ or (λ =) v / f or (λ) = 1500 / 13 000
1500 / 13 000 or 1.2 × 10N
(c) 330 m / s ≤ speed ≤ 350 m / s

Q143 • Oct/Nov 2024 Paper 41 Question 5 — 3.3 Sound and ultrasound

(a) 0.20 m
(b) any value in range from 330 m / s ≤ value ≤ 350 m / s
(c)(i) (b) ÷ (a) evaluated and Hz
f = v / λ or (f =) v / λ or (b) ÷ (a)
(c)(ii) audible/yes/it is or inaudible / no / it isn't consistent with value in 5(c)(i)
consistent explanation with reference to 20 (Hz) ≤ normal range of human hearing ≤ 20 000 (Hz)
(d) 1 (explanation mentions) diffraction
2 Only a little diffraction
3 (because) gap width large (compared to wavelength)
4 Little / no sound heard at J and (some) sound heard at K

Q144 • Oct/Nov 2024 Paper 42 Question 6 — 3.3 Sound and ultrasound

(a)(i) diffraction
(a)(ii) 450 Hz
v = f λ or (f =) v / λ or 340 / 0.75
(a)(iii) large diffraction when gap size / doorway is similar to wavelength
high frequency / 3500 Hz has (much) shorter wavelength and there is less diffraction (with shorter wavelengths)
(b) ray drawn from the violin to point X and from point X to the teacher
mirror drawn at point X and labelled and angle of incidence = angle of reflection
correct arrow on incident ray or correct arrow on reflected ray

Q145 • Oct/Nov 2024 Paper 42 Question 8 — 3.3 Sound and ultrasound

(a) ultrasound

(b) 0.029 s

(distance travelled =) 22 × 2 or 44 SEEN

v = s / t or (t =) s / v or 44 / 1500

(c) reflected wave is weaker / has smaller amplitude

fish is small(er); or only small part of wave reflects off fish

Q146 • Feb/March 2025 Paper 42 Question 7 — 3.3 Sound and ultrasound

(a)(i) diffraction

(a)(ii) 4.8 × 10-7 m or 480 nm

One wavelength marked on Fig. 6.1 or 1.2 seen

(a)(iii) at least two curved wavefronts

three semi-circular wavefronts centred on (centre of) gap

wavelength is unchanged

(b)(i) v = fλ

(λ =) 330 / 380 or (λ =) 0.87 (m)

(b)(ii) 7.6 s

v = s / t or (t =) s / v or (t =) 2500 / 330

Q147 • May/June 2025 Paper 42 Question 5 — 3.3 Sound and ultrasound

(a) echo method (outside):

1 method of producing short, loud sound (e.g. clap / shout / cry out / bang 2 pieces of wood together)

2 measuring tape or trundle wheel, stopwatch and wall

3 (person) starts stopwatch when clap etc heard and stops it when echo heard

4 speed = 2 × measured distance ÷ time

direct method outside:

1 method of producing short, loud sound (e.g. fire gun)

2 measuring tape or trundle wheel, stopwatch

3 (student at one end) starts stopwatch when smoke seen from gun and stops it when sound heard

4 speed = distance ÷ time

direct method (using digital):

1 method of producing short sound e.g. clap or hammer striking block

2 measuring tape or metre ruler, digital timer and microphones

3 digital timer starts when sound reaches first microphone and stops when sound reaches second microphone. Time difference is recorded on digital timer

4 speed = distance ÷ time for direct method

(b)(i) radio waves

(b)(ii) 3(.0) × 108 m / s

(b)(iii) 0.12 m

v = fλ or (λ =) v ÷ f or (λ =) {3(.0) × 108} ÷ {2.48 × 109}

(λ =) {3(.0) × 108} ÷ {2.48 × 109} or (λ =) 1.2 × 10N

Q148 • May/June 2025 Paper 43 Question 6 — 3.3 Sound and ultrasound

(a) sound with a frequency higher than 20 kHz

(b) vibrations (of the wave / particles) are parallel to the direction of propagation

(c)(i) (pulse of) ultrasound / sound / wave (sent into water) reflects from object

time to travel to object and back measured

depth = speed × time

(c)(ii) Any one valid point(s):

• non-destructive testing of materials

• medical scanning (of soft tissue)

Q149 • Oct/Nov 2025 Paper 41 Question 5 — 3.3 Sound and ultrasound

(a) (speed of sound in air =) 330 m / s ≤ value ≤ 350 m / s
(speed of sound in water) is faster
(b) Yes and (normal) human hearing range is 20 Hz to 20 kHz (and all dolphin sounds lie in this range)
Yes and all the dolphin sounds are in the range of human hearing
(c)
Frequency kHz amplitude loudness pitch
14 Large loud high
8 small soft / quiet low
one mark for each column correct
(d) Particles
(d) high pressure
low pressure
Parallel

3.4 Light, refraction and lenses

Q150 • Feb/March 2021 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a) Any two correct rays from
• from O through optical centre (and beyond)
• from O parallel to principal axis to centre line of lens then through F1
• from F2 through O to centreline of lens then parallel to principal axis
rays traced back to intersect and 2.4 - 3.6 cm
(b) magnified
same way up as object
virtual
(c) one ray from each prism refracted towards principal axis
(rays) converge to the right of original convergence on the principal axis

Q151 • May/June 2021 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a) blue ray refracted MORE towards normal at first surface
refraction away from normal at second surface
ray of blue light below ray of green light and diverging throughout path (after entering prism)
(b) v = fλ in any form or (f=) v / λ
(f =) 3 × 108 ÷ 4.8 × 10-7
(f =) 6.3 × 1014 Hz

Q152 • May/June 2021 Paper 43 Question 6 — 3.4 Light, refraction and lenses

(a) principal focuses marked in correct position
(b) 1 mark for each of:
• 1 correct ray
• 2nd correct ray
• 3rd correct ray and image, labelled I, in correct position with arrow at bottom
(c) real
inverted and enlarged
(d)(i) (image produced by a magnifying glass is) upright
or NOT inverted or virtual
(d)(ii) position marked between principal focus and lens

Q153 • Oct/Nov 2021 Paper 41 Question 5 — 3.4 Light, refraction and lenses

(a) (point) where (parallel) rays (of light) meet (after passing through lens)
Key physics: parallel rays light meet / focussed after passing through lens
that emerge parallel pass (before reaching lens)
(b) distance between principal focus / focal point and optical centre / lens
(c)(i) vertical line labelled L 4.0 (± 0.2) cm to the right of O
(c)(ii) Key physics: paraxial ray tip O candidate's lens I
to candidate's lens
3.0 (± 0.2) cm
(c)(iii) fourth box ticked i.e:
reversed / inverted

Q154 • Oct/Nov 2021 Paper 42 Question 7 — 3.4 Light, refraction and lenses

(a)(i) i = 60° used or seen
sin i / sin r =n in any form
ray refracted toward normal and toward AC
ray clearly refracted down in prism reaching AC with r = 35(°)
(a)(ii) 10°
(b) refracted away from normal
(c)(i) (total internal) reflection at X
NOT refraction at X or anywhere else
reaches end of fibre with only one additional reflection (off lower internal edge of fibre)
(c)(ii) total internal reflection

Q155 • Oct/Nov 2021 Paper 43 Question 7 — 3.4 Light, refraction and lenses

(a)(i)
ray approaching left hand face of prism closer to normal than emerging ray
ray entering right hand face of prism showing refraction towards normal for ray already drawn
(a)(ii) light of single frequency
(b)(i) 3(.0) × 108 m / s
(b)(ii) 5.8 × 1014 Hz
(f =) v / λ in any form or 3.0 × 108 / 5.2 × 10-7
(b)(iii) 2.0 × 108 m / s
refractive index = speed of light in air / speed of light in glass in any form

Q156 • May/June 2022 Paper 41 Question 7 — 3.4 Light, refraction and lenses

(a) (all the light) meets (at a point) or is focused or intersects
(all the light) travels towards a point
it then diverges; or spreads out (from that point)
3.0 cm from lens
(b) two marked points on XY 3.0 cm from centre of lens and one on left and one on right and each labelled F
(c)(i) two of these rays from tip of N drawn:
ray (that seems to come) from left-hand principal focus and emerges from lens paraxially
paraxial ray to lens and then towards right-hand principal focus
ray towards / through centre of lens
two rays traced back to intersection
and line from intersection to axis and line labelled I
(c)(ii) virtual and light / rays do not pass through I or
virtual and light / rays only seem to come from I or
virtual and produced by diverging rays
virtual and (real) rays do not meet
(c)(iii) magnifying glass

Q157 • May/June 2022 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a) 1.9-2.1 cm

(b) (circle round) enlarged

(circle round) inverted

(circle round) real

(c) not an intersection of rays; or cannot be formed on a screen

pass through image or light rays do not meet or light rays do not converge

Q158 • Oct/Nov 2022 Paper 41 Question 6 — 3.4 Light, refraction and lenses

(a) (light of a) single frequency

(b)(i)

angle of incidence is 0°

(hence) angle of refraction is 0°

all the wavefront hits the plastic at the same time

all slows down at the same time

(b)(ii) 1.8 × 108 m / s

n = 1 / sin c in any form or n = 1 / sin 37°

(n =) 1.7

vpl = v0 / n in any form or 3.0 × 108 / 1.7 or 3.0 × 108 × sin 37°

(b)(iii)

critical angle (for blue light) < 37° or critical angle for red (light) is 37°

angle of incidence (of blue light) greater than its critical angle (in plastic)

total internal reflection or all the (blue) light reflects or no (blue) light

leaves the glass / refracts / travels in air along the straight edge

Q159 • Oct/Nov 2022 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a)(i) two correct rays from:

• ray from X through centre of lens

• ray from X to lens, parallel to principal axis, refracted through RH focus F

• ray from X (that would pass through LH focus) refracted parallel to principal axis.

two rays correctly extended back, intersecting to left of object and image labelled

IY drawn and 36 mm ≤ distance ≤ 44 mm

(a)(ii) Any two valid point(s):

• object closer to lens than (one) focal length

• (actual) rays do not meet (at image)

• image cannot be formed on a screen or image only visible through lens

• object and image on same side (of lens) or image on LHS of lens/object.

(b)

blue ray refracted closer to the normal than the green ray as it enters the prism

blue ray refracted away from the normal as it leaves the prism

Q160 • Oct/Nov 2022 Paper 43 Question 7 — 3.4 Light, refraction and lenses

(a) Any two valid point(s):

• all light is reflected

• no light is refracted

• (occurs) when light travels in a more dense medium towards a (boundary with a) less dense medium

(b)(i) (x =) 48°

n = 1 / sin c or c = sin-1 (1 / n) or sin c = 1 / 1.5 or c = sin-1 (1 / 1.5)

c = 42(°)

(b)(ii) (speed =) 2.0 × 108 m / s

speed of light in vacuum (approx.) speed of light in air

n= or n =
speed of light in liquid speed of light in liquid
or n = c / v or (v =) c / n
3×108
or 1.5 =
speed of light in liquid

Q161 • Feb/March 2023 Paper 42 Question 5 — 3.4 Light, refraction and lenses

(a)(i) 42°
n = 1 / sin c or c = sin-1 (1 / n) or c = sin-1 (1 / 1.5)
(a)(ii) ray continues along radius of semicircle within plastic
ray reflected inside plastic on straight edge, with angle of reflection = angle of incidence and emerges from block along the normal
(b)(i) (focal length =) 7.2 cm
(b)(ii) two correct rays from:
• ray from top of object through centre of lens
• ray from top of object (that would pass through F on LHS of lens) refracted parallel to the principal axis
• Key physics: ray top object lens parallel principal axis refracted through F same distance right as marked on left of lens)
Key physics: rays correctly extended back intersect left object line principal axis top image labelled I.
(c) diverging lens in front of eye lens
rays meeting on the retina

Q162 • May/June 2023 Paper 41 Question 5 — 3.4 Light, refraction and lenses

(a) (light / electromagnetic radiation) of a single frequency
(b) angle of incidence / i = 0 or incident ray along normal
or all of wavefront enters block at same time
angle of refraction / r = 0 or no refraction
or whole wavefront slows down at same time
(c) (c =) sin-1{1 /1.5} (= 42°)
(c =) sin-1 {1 / n} = 41.8°
n = 1 / sin c or (c =) sin-1 {1 / n} or (c =) 41.8°
(d)(i) all light is reflected
θ / angle of incidence > c / critical angle
(d)(ii) all light is reflected or reflected ray at 90° to incident ray or reflected ray is parallel to original ray

Q163 • May/June 2023 Paper 42 Question 7 — 3.4 Light, refraction and lenses

(a) normal drawn in correct position and at right angles to the surface
(b) 22°
i = 34(°)
n = sin i / sin r or (r =) sin-1 {sin i / n} or sin r = sin 34 / 1.47 or sin r = 0.38
(c) 3.0 × 108 m / s
(d) 2.04 × 108 m / s
n = speed of light in air / speed of light in oil
or (speed of light in oil =) speed of light in air / n
or (speed of light in oil =) 3.0 × 108 / 1.47

Q164 • May/June 2023 Paper 43 Question 4 — 3.4 Light, refraction and lenses

(a)(i) two correct rays extended back (towards the intersection)
extended rays intersect
image drawn from intersection to principal axis and base of image lies in correct range
(a)(ii) circles around:

- enlarged
- virtual
- upright

(b)(i) long-sightedness / long sight / far-sighted

(b)(ii) converging lens reduces focal length of eye; or converging lens brings focal point forward

behind back of eye

(so that) rays converge / focus on back of eye / retina

Q165 • Oct/Nov 2023 Paper 41 Question 6 — 3.4 Light, refraction and lenses

(a)(i) Any two valid point(s):

- ray from top / bottom of object, parallel to principal axis, refracted through right-hand principal focus
- straight ray from same point on object through optical centre
- ray that (seems to) come from left-hand principal focus through same point of object and refracted parallel to principal axis

rays traced back to intersection and intersection / image labelled I

(a)(ii) (distance =) 35.5 cm to 38.5 cm

7.1 to 7.7 (cm) or (distance =) 35.0 (cm) to 40.0 (cm)

(b) virtual and Any one valid point(s):

- cannot be projected on a screen
- (real) light (ray) does not pass through image
- light only seems to come from image

(c) Any one valid point(s):

- long-sightedness focuses image behind retina / back of eye; or longsightedness produces blurry / fuzzy images (of close objects)
- converging lens reduces focal length (of eye)
- (converging lens) puts image further away (from the eye)

Key physics: converging lens gives sharp/focussed image retina / back eye rays converge

Q166 • Oct/Nov 2023 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a) (p.d. across LED = 4.5 - 1.2 =) 3.3 V

(V =) IR

(b) Key physics: LED diode only allows current direction / has very high resistance

reversed.) or (it) is reverse-biased

(c) E=IVt or (t =) E / VI or Q = E / V and Q = I × t

(t =) 1050 ÷ [0.02 × 4.5 × 3600] or (t =) 3.2 h

(d) (charge =) 72 C

I =Q / t or (Q =) It or (Q =) 0.02(0) × 3600

Q167 • Feb/March 2024 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a) ray from top of object to tip of image

line labelled L drawn perpendicular to principal axis at its intersection with previous ray

(b) ray from top of O parallel to principal axis to lens and ray from lens to tip of I

or ray from tip of I parallel to principal axis to lens and ray from lens to top of O

2.1 cm

(c) virtual

upright

Q168 • May/June 2024 Paper 41 Question 4 — 3.4 Light, refraction and lenses

(a)(i) (point on principal axis) where rays of light parallel (to the principal axis, incident on converging lens)

(rays) meet / converge after passing through lens / refraction

(a)(ii) X marked between one of the focal points and the lens and E marked on other side of lens

(a)(iii) virtual and upright
(b)(i) 2.0 × 108 m / s
n = c / vg or (vg =) c / n or (vg =) 3(.0) × 108 / 1.5
(b)(ii) (wavelength) decreases
(c) long-sightedness
Key physics: moves image towards lens / back eye retina reduces shortens focal length combined (converging lens) focuses image on back of eye / retina

Q169 • May/June 2024 Paper 42 Question 5
3.4 Light, refraction and lenses

(a) ray travels along the normal or angle of incidence = 0(°)
(b)(i) n = 1 / sin c or (n =) 1 / sin c or (n =) 1 / sin 42 (°)
1.5
(b)(ii) ray reflected at BC and no refracted ray
ray hits AC with angle of incidence = 0 °
correct refraction of candidate's ray into air at AC
(c) Any two valid point(s):
• high rates (of data transmission) / fast (data transmission)
• carry large amounts (of data / information)
• secure
• little signal / data loss
• glass is transparent to (some) infrared

Q170 • May/June 2024 Paper 43 Question 6
3.4 Light, refraction and lenses

(a) where rays of light parallel (to the principal axis) converge after passing through lens
(focal length is) the distance between (centre of) the lens and principal focus
(b) Any two valid point(s):
• ray from top of object to lens, parallel to principal axis, refracted to F2
• ray from top of object through centre of lens, undeviated
• ray from F1, through top of object and on to lens, then parallel to principal axis
rays extrapolated back to converge to the left of F1
image drawn from principal axis to intersection with arrow (to show orientation)

Q171 • Oct/Nov 2024 Paper 43 Question 4
3.4 Light, refraction and lenses

(a)(i) any two correct = 1 mark
all three correct = 2 marks
• speed decreases / gets less / slower
• wavelength decreases / gets smaller / shorter
• frequency unchanged / stays the same / no effect
(a)(ii) (n =) sin 45 / sin 30
(a)(iii) 46° or 45°
n = 1 / sin c or (c =) sin-1 (1 / n) or (c =) sin-1 (1 / 1.4) or sin c = 1 / n
(b)(i) total internal reflection and i = r for initial reflection
ray reaches end of fibre after a total of 2 or 3 reflections only
(b)(ii) Any two valid point(s):
• internet transmission / (highspeed) broadband
• telephone networks or telecommunications
• cable TV
• endoscope
• lasers in surgery
• microscopy
• military aircraft wiring
• imaging (cameras) in industry

- inspection of pipes or other hard to reach places

Q172 • Feb/March 2025 Paper 42 Question 6 3.4 Light, refraction and lenses

(a)(i) Horizontal axis labelled P

(a)(ii) X on horizontal axis 3.0 cm to the left of centre of L or X on horizontal axis 3.0 cm to the right of centre of L

(a)(iii) Any two valid point(s):

- Straight line from a point on O, passing through centre of L (and beyond)
- Horizontal line from same point on O to L, refracted through F (on RH side of L)
- Key physics: Straight line same O through F LH side L refracted parallel principal axis

both rays extended until they intersect

(a)(iv)

diminished
enlarged
inverted
real
same size
upright
virtual

(b) (image is now) virtual

(image is now) upright

Q173 • May/June 2025 Paper 41 Question 5 3.4 Light, refraction and lenses

(a) (refractive index is) the ratio of the speed of light in two different regions

speed of light in air
or (refractive index =)
speed of light in (soap) film

(b)(i) sin i / sin i / sin 60

n= or (r =) sin-1 ( ) or 1.28 =
sin r / n / sin r
i = 60 (°)

(b)(ii) normal drawn (at the point incident ray meets film)

refracted ray drawn (refraction towards normal in film) and angle of refraction labelled
braille: angle identified but not labelled

(c)(i) (light of) a single frequency

(c)(ii) 4.4 × 1014 Hz

(speed of light / e-m waves is approximately) 3.0 × 108 m / s (in air)
v = fλ or (f =) 3.0 × 108 ÷ 680 × 10-9

Q174 • May/June 2025 Paper 42 Question 6 3.4 Light, refraction and lenses

(a)(i) two points labelled F on principal axis, both 3.0 cm from the centre of the lens

(a)(ii) method 1

two correct rays from:

- ray from top of I through the centre of the lens
- Key physics: ray top I parallel principal axis left lens centre then through RH focus
- Key physics: ray top I through LH focus lens centre then parallel principal axis right

method 2
two correct rays from:

- ray from top of I through the centre of the lens
- Key physics: ray top I through RH focus lens extended back centre parallel principal axis left the lens
- Key physics: ray top I parallel principal axis left lens extended back centre through

the LH focus
object, labelled O, at the intersection of rays to the RH of the lens for method 1
object, labelled O, at the intersection of rays to the LH of the lens for method 2
(a)(iii) real or inverted
(b)(i) rays meeting behind the retina or would meet behind retina if extended
(b)(ii) any converging lens

Q175 • Oct/Nov 2025 Paper 41 Question 6 — 3.4 Light, refraction and lenses

(a) Any two valid point(s):
• high rates (of data transmission) / faster (data transmission)
• carry large amounts (of data / information)
• secure
• Little data / signal loss
• glass is transparent to (some) infrared
(b)(i) (c =) 46°
N = 1 / sin c, 1.4 = 1 / sin c or (c =) sin-1 (1 / 1.4)
Equation in this form only
(c =) sin-1 (1 / n) or (c =) sin-1 (1 / 1.4) or (c =) 46
(b)(ii) angle of incidence (of light) at which the angle of refraction is (exactly) 90°
(b)(iii) angle of incidence marked and
TIR of ray inside glass and light ray leaves end of fibre after two; or three reflections
angle of incidence marked or TIR drawn until ray leaves end of fibre

Q176 • Oct/Nov 2025 Paper 42 Question 6 — 3.4 Light, refraction and lenses

(a)(i) wavefronts joined at the boundary
angle with the surface in region B smaller than angle with surface in region A and slanting in correct direction
refracted wavefronts all parallel slanting in the correct direction and wavelength less than incident waves
(a)(ii) refraction
(b)(i) speed of light in first region is less than speed of light in second region
angle of incidence must be greater than the critical angle
(b)(ii) Any two valid point(s):
• high rates of data (transmission)
• carry large amounts of data / information
• secure
• little data / signal loss
• glass is transparent to visible light and (some) infrared

3.5 Electromagnetic spectrum

Q177 • Oct/Nov 2021 Paper 41 Question 6 — 3.5 Electromagnetic spectrum

(a)(i) (J) ultraviolet (radiation)
(K) infrared (radiation)
(L) radio (waves)
two correct
all three correct
(a)(ii) L or radio (waves)
(b) (c =) 3.0 × 108 (m / s) seen
(f =) v / λ in any form or 3.0 × 108 / 1.2 × 10-9
2.5 × 1017 Hz
(c)(i) stated medical use (e.g. treating cancer / X-ray shadowgraph / sterilising equipment)
statement of what happens to the X-rays (e.g. absorbed by tumour / bones / bacteria)

stated consequence (e.g. tumour killed or image / picture / shadow / photograph produced)

(c)(ii) can cause burns / (cell) mutation / cell damage / tumours / cancer / damages DNA etc.

Q178 • May/June 2023 Paper 41 Question 6 3.5 Electromagnetic spectrum

(a) (wavelength =) 0.16 m

v = fλ or (λ =) v / f or (λ =) 3 × 108 / 1.9 × 109

(b) (microwaves) only need short aerials / antennas

(microwaves) penetrate (some) walls

(c)(i) labelled diagram of digital (signal) with blocks of high (1) and low (0) and labelled diagram of analogue with continuously

variable signal

digital (signal) consists of two values

analogue (signal) varies over a range (of values)

(c)(ii) Any two valid point(s):

• faster (data) transmission rate or data can be compressed

• data / signal transmitted over long(er) distances (as signal can be regenerated)

• noise easily removed (from signal / data) or signal can be regenerated

Q179 • May/June 2023 Paper 43 Question 5 3.5 Electromagnetic spectrum

(a)(i) infrared

(a)(ii) glass is transparent to visible light and (some) IR

(visible light and some IR) can carry high rates of data / information

(b)(i) (n =) 1.4

(n =) 1 / sin c or (n =) 1 / sin c or (n =) 1 / sin 45

(b)(ii) diagram shows that total internal reflection is taking place inside optical fibre

one or more correct reflections with i > 45°

Q180 • Oct/Nov 2023 Paper 41 Question 4 3.5 Electromagnetic spectrum

(a)(i) (point / place / position) where (all) the weight (seems to) acts

(a)(ii) Key physics: small tilt / rotation makes G no longer vertically above base produces moment topples

transmitter)

(b)(i) arrow(head) marked along wire W towards ground

(b)(ii) moment = F × d and correct indication of F and d on Fig. 4.1.

(moment is) force × (perpendicular) distance (from base / pivot)

(moment is) force × perpendicular distance (from base / pivot)

(c) a use of radio waves, e.g. RFID / astronomy / Bluetooth / RADAR / wifi

Q181 • Feb/March 2024 Paper 42 Question 5 3.5 Electromagnetic spectrum

(a)(i)

application region of electromagnetic spectrum

cancer treatment gamma rays

bluetooth radio waves

optical fibres infrared

security marking ultraviolet

sterilising food gamma rays

wireless internet Microwaves

(a)(ii) 3.0 × 108 (m / s) or 300 000 000 (m / s)

(b)(i) three crests parallel to the barrier

same wavelength as wave after the gap

(b)(ii) central part of crest (parallel to the (gap in the) barrier) is straight

crests have curved ends

Q182 • Oct/Nov 2025 Paper 43 Question 5 — 3.5 Electromagnetic spectrum

(a)(i) all four correct = 2 marks; any two correct = 1 mark

security marking gamma rays

Bluetooth ultraviolet

optical fibres infrared

detection of cancer radio waves

(a)(ii) Any one valid point(s):

- only require a short aerial / antenna
- can pass through the atmosphere / ionosphere to satellites
- can transmit large amounts of data at high speeds / high rate of data transmission

(b) Any two valid point(s):

transverse:

- oscillations / vibrations are at right angles to direction of propagation
- medium not required / can travel through a vacuum (for e-m wave)
- have crests / peaks and troughs

longitudinal:

- oscillations / vibrations are parallel to direction of propagation
- medium is required
- have rarefactions and compressions

(c)(i) longitudinal

(c)(ii) refraction

Any one valid point(s):

- change in speed of the wave
- change in density of the medium

(d) increase in carbon dioxide / greenhouse gases (in the atmosphere)

increase in methane / water vapour (in the atmosphere)

Any one valid point(s):

prevents / reduces (re-emitted) radiation escaping (into space)

radiation re-emitted / reflected back to surface (by atmosphere / greenhouse gases)

(amount of) radiation absorbed is greater than (the amount of) radiation leaving the atmosphere

3.6 Waves - mixed

Q183 • Feb/March 2022 Paper 42 Question 5 — 3.6 Waves - mixed

(a)(i) straight line from clock to mirror and from mirror to eye with correct arrows

Angle of incidence = angle of reflection

(a)(ii) Correct position of image

(a)(iii) Virtual (no mark)

Cannot be projected on a screen / light doesn't pass through image / AW

(a)(iv)

(b)(i) (monochromatic light) is light of a single frequency

(b)(ii) 5.4 × 1014 Hz

(speed of light =) 3 × 108 (m / s)

(f =) v / λ in any form or 3.0 × 108 / 5.6 × 10-7

Q184 • May/June 2025 Paper 43 Question 5 — 3.6 Waves - mixed

(a) (refractive index is) the ratio of the speed of light in two different mediums

speed of light in air

or (refractive index =)

speed of light in water

sine of angle of incidence

or (refractive index =)
sine of angle of refraction

(b) P continues vertical

Q refracted away from normal in the correct direction

(c) 1.3

(n =) 1 ÷ sin 49 or (n =) 1 ÷ sin c

(d) Any two valid point(s):

• (ray travelling from) dense to less dense medium or water is more dense than air
• critical angle = 49°
• angle of incidence exceeds critical angle / 49°

3.7 Medical imaging and electromagnetic waves

Q185 • May/June 2024 Paper 43 Question 8 3.7 Medical imaging and electromagnetic waves

(a)(i) (ultrasound is) sound with a frequency higher than 20 kHz

(a)(ii) (sound travels) faster (in a liquid than in air) /

(a)(iii) 3.0 × 108 m / s

(b) Any three valid point(s):

• X-rays and ultrasound show internal body parts (without the need to cut open the body)
• X-rays and ultrasound both travel through (some parts of) the body
• images are formed by ultrasound (partially) reflecting (from boundaries between body matter)
• images are formed by X-rays which (travel in straight lines and) pass through soft tissue and are absorbed by bone
• X-rays show parts of the skeleton / X-rays show bones; or ultrasound shows image of soft tissues
• an X-ray detector forms image from X-rays that travel through patient; or ultrasound detector forms image from sound waves reflected back (from body parts)
• X-rays expose you to radiation (which can be harmful to humans)